

Shenglong Chen

Vancouver, BC, Canada | Open to Relocation

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Software engineer with experience designing and building backend services, distributed data pipelines, and cloud-based systems. Strong foundation in concurrency, database design, and performance optimization. Experienced in Agile development environments and rapid prototyping using modern AI-assisted engineering workflows.

Technical Skills

Languages & Frameworks: Python (FastAPI, Flask, Pandas, NumPy), Java (Spring Boot, JUnit 5), SQL (PostgreSQL, MySQL), R (Statistical Modeling), JavaScript (React/Next.js)

Backend & Systems: FastAPI, Flask, Spring Boot, REST APIs, concurrency, multithreading/multiprocessing, distributed systems

Cloud & Infrastructure: AWS (Lambda, S3, RDS, EC2), Docker, Linux/Unix, GitHub Actions (CI/CD)

Databases: PostgreSQL, MySQL, schema design, query optimization, indexing

Development Practices: Agile, Kanban, code reviews, CI/CD, test-driven development

AI-Assisted Development: Claude (code generation/debugging), Git worktrees (parallel development), structured prompt workflows, MCP-based tool integration (experience)

Education

The University of British Columbia

Vancouver, BC

Bachelor of Science, Major in Cognitive Systems (Artificial Intelligence)

Sep 2022 – Dec 2025

○ **Relevant Coursework:** Distributed Systems, Advanced Data Structures, Algorithm Design, Databases, Computer Systems, Artificial Intelligence, Statistical Modeling

Professional Experience

Simon Fraser University

Burnaby, BC

Machine Learning Engineering Intern (Software Engineering)

May 2025 – Oct 2025

- Designed and implemented backend data pipelines in Python to ingest, clean, and transform 10K+ financial records, improving throughput by **4x** using multiprocessing and optimized I/O handling
- Designed relational database schemas and optimized PostgreSQL queries through indexing and execution plan analysis; conducted load testing to identify bottlenecks, reducing execution time by **35%** and improving stability under concurrent workloads
- Integrated data processing into production workflows for reliable automated analysis
- Participated in Agile development cycles, contributing to API design, code reviews, and iterative feature delivery using Git

Key Technical Projects

Asynchronous Job Processing System: [Python / FastAPI / Redis / PostgreSQL / Docker / k6] 12/2025 – 02/2026

- Built a distributed backend system for asynchronous job execution using Redis-based priority queues and worker pools
- Identified queue and database bottlenecks under load testing; optimized worker parallelism and connection handling, improving throughput and system stability under concurrent workloads
- Achieved **150–300 jobs/sec throughput** under load testing with **500–1,500 concurrent requests** (k6)
- Implemented reliability mechanisms including retry logic, idempotent job handling, and dead-letter queues to handle failure scenarios
- Introduced rate limiting and priority scheduling to control load and ensure fair resource allocation
- Instrumented logging and metrics to track queue depth, job latency, and system performance in real time

Distributed Task Microservice: [Python / FastAPI / PostgreSQL / Docker] 01/2025 – 03/2025

- Designed and implemented a modular backend service with RESTful APIs supporting authentication (JWT), task management, and persistent storage
- Structured service into layered components (API, service, data access) to improve maintainability and separation of concerns
- Optimized database interactions by eliminating N+1 queries and improving indexing strategy, reducing response latency under **200+ concurrent requests**
- Built CI/CD pipelines using GitHub Actions; incorporated AI-assisted workflows to accelerate debugging and test generation

Event-Driven Data Processing Pipeline: [Python / AWS Lambda / S3 / Docker / k6 / Locust] 06/2024 – 09/2024

- Designed and implemented a serverless pipeline to process **1M+ events/day** using AWS Lambda and S3 with high availability
- Simulated **5,000+ concurrent users** using k6 and Locust, identifying database connection leaks and resource bottlenecks
- Reduced system latency by **40%** through connection pooling and optimized resource allocation
- Implemented structured logging and monitoring to detect anomalies and support debugging in distributed environments

AI Heuristic Solver Engine: [Java] 01/2024 – 04/2024

- Implemented a high-performance solver using heuristic search and pruning techniques to reduce search space
- Achieved **60% reduction in runtime** and **45% reduction in memory usage** through optimized data structures and caching
- Designed system using Strategy and Observer patterns to support extensibility and modular architecture